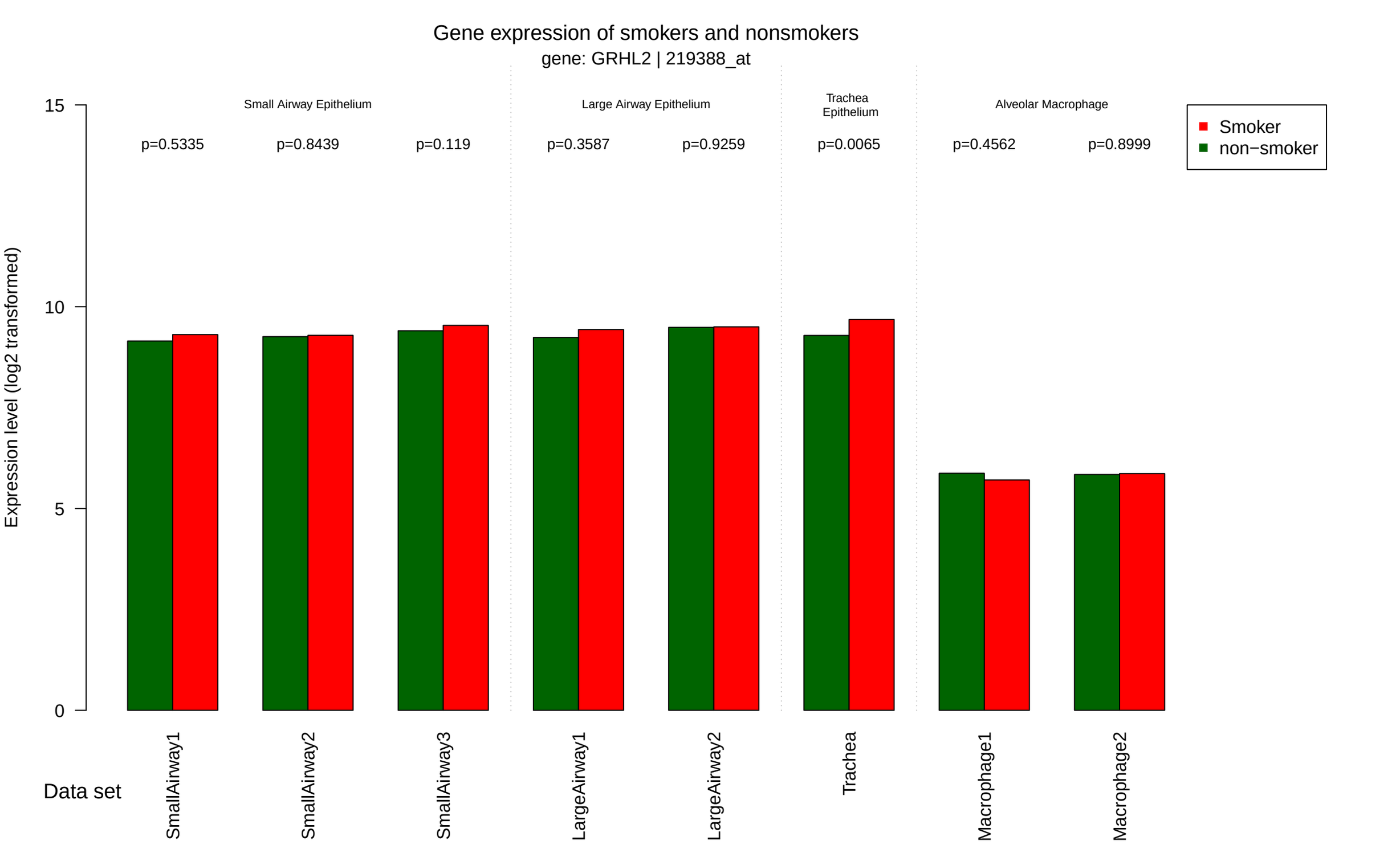
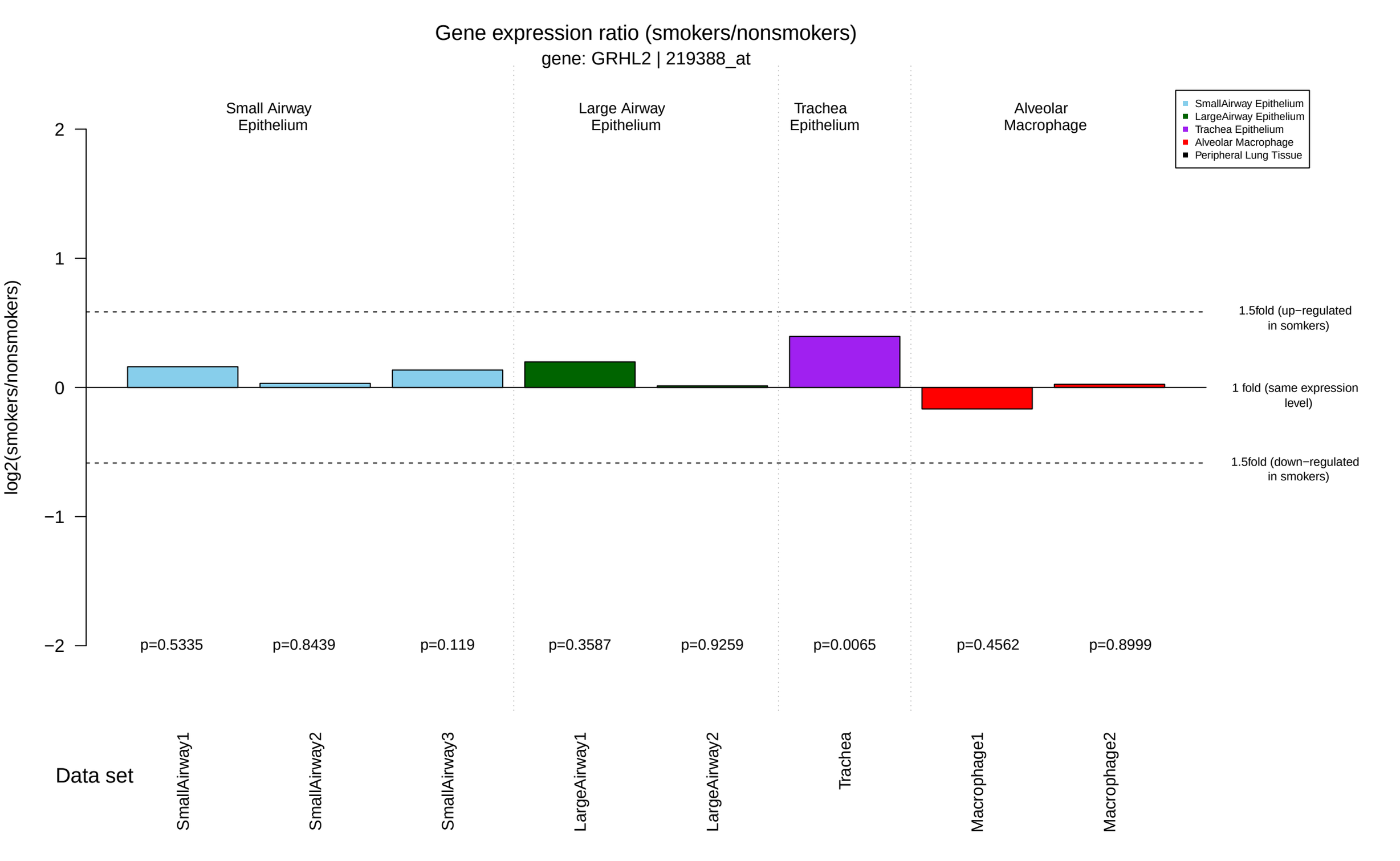
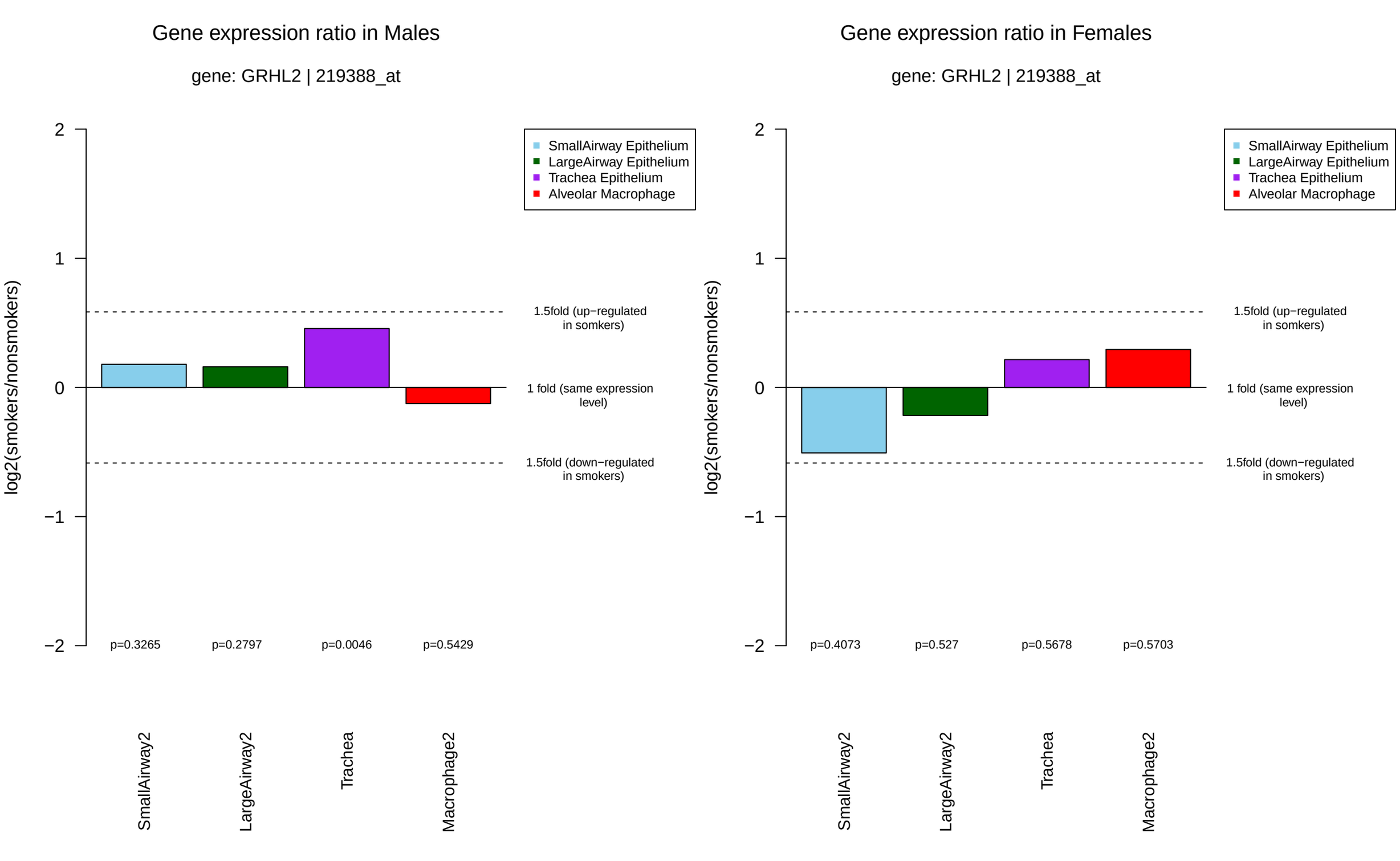
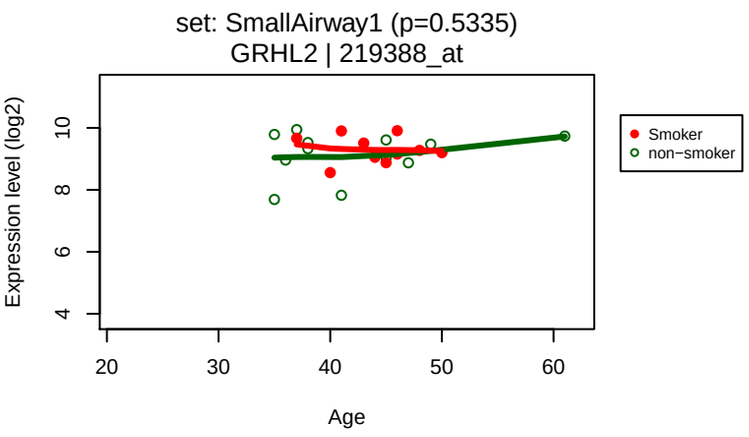
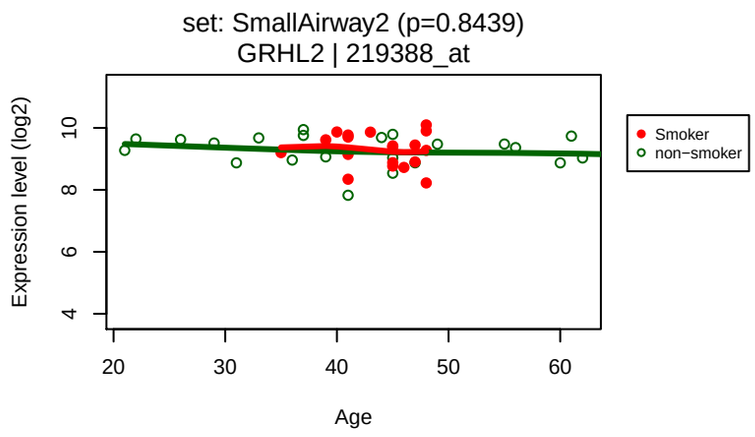
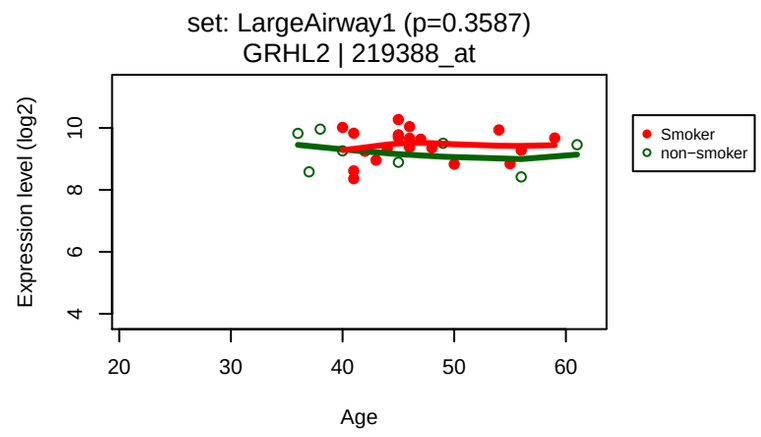

Gene expression of smokers and nonsmokers
gene: GRHL2 | 219388_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5335
p=0.8439
p=0.119
p=0.3587
p=0.9259
p=0.0065
p=0.4562
p=0.8999
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

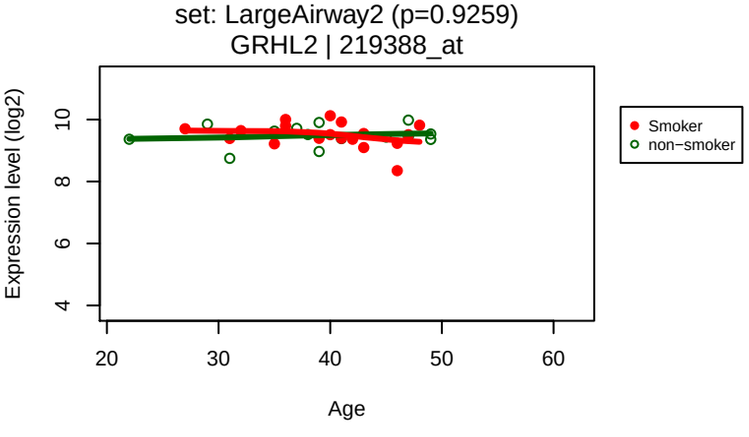

Gene expression ratio (smokers/nonsmokers)
gene: GRHL2 | 219388_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5335
p=0.8439
p=0.119
p=0.3587
p=0.9259
p=0.0065
p=0.4562
p=0.8999
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

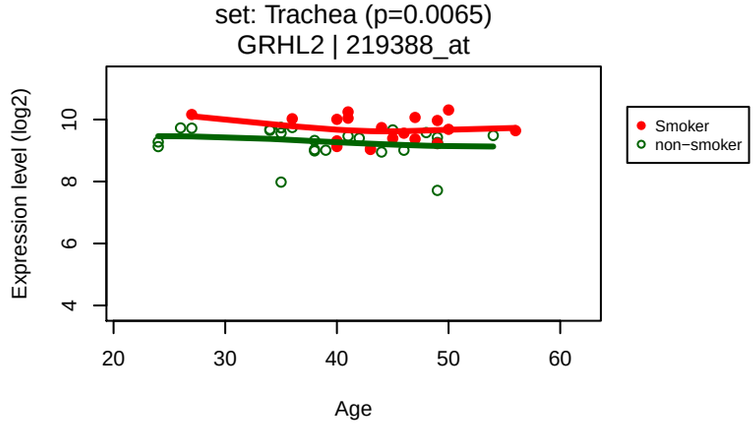

Gene expression ratio in Males
gene: GRHL2 | 219388_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3265
p=0.2797
p=0.0046
p=0.5429
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRHL2 | 219388_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4073
p=0.527
p=0.5678
p=0.5703
SmallAirway2
LargeAirway2
Trachea
Macrophage2

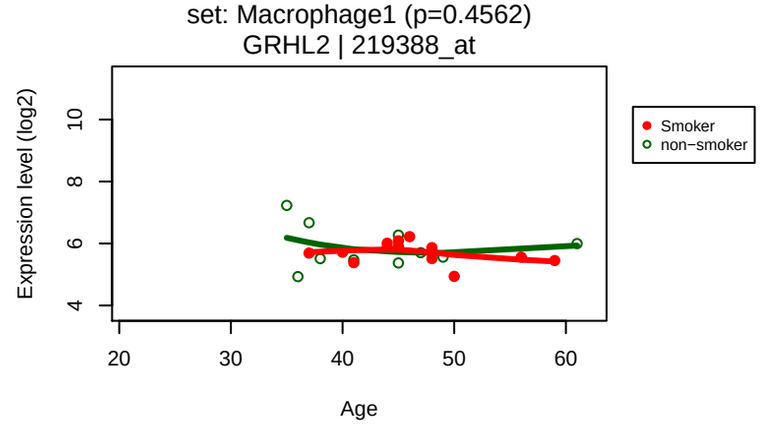

set: SmallAirway1 (p=0.5335)
GRHL2 | 219388_at
Expression level (log2)
Age
Smoker
non−smoker

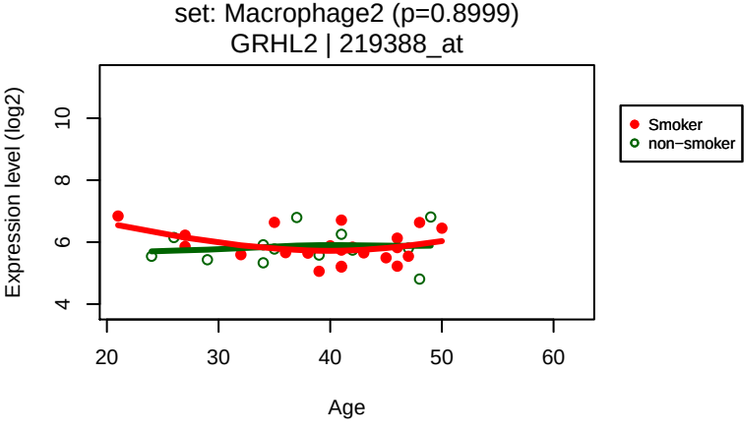

set: SmallAirway2 (p=0.8439)
GRHL2 | 219388_at
Expression level (log2)
Age
Smoker
non−smoker

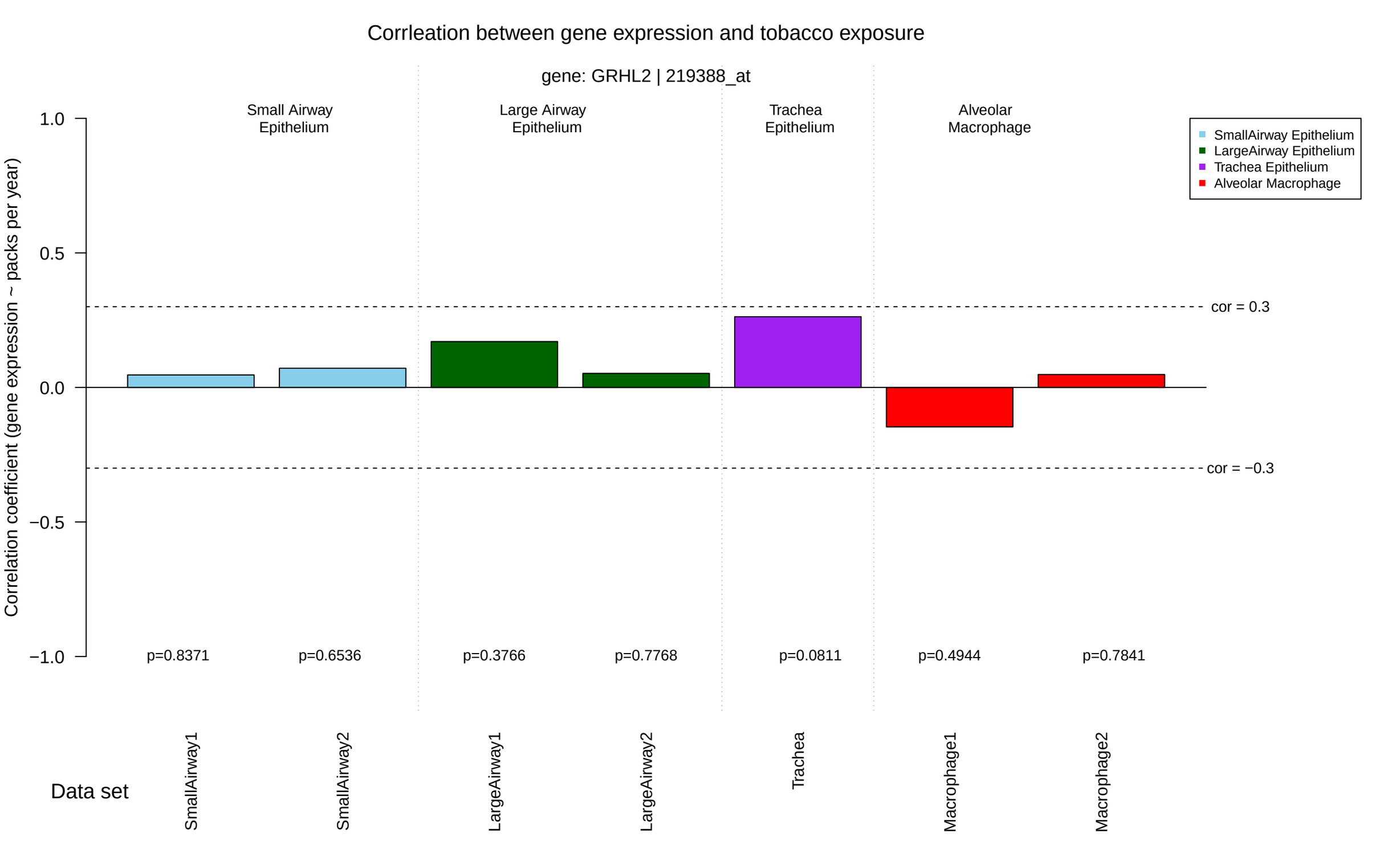

set: LargeAirway1 (p=0.3587)
GRHL2 | 219388_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9259)
GRHL2 | 219388_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0065)
GRHL2 | 219388_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4562)
GRHL2 | 219388_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8999)
GRHL2 | 219388_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRHL2 | 219388_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8371
p=0.6536
p=0.3766
p=0.7768
p=0.0811
p=0.4944
p=0.7841
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2